Industrial Internship Report on

”FOOD DELIVERY APPLICATION: TESTYBITES”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was the development of "TestyBites," a comprehensive Android-based Food Delivery Application. It features a complete ecosystem with an Admin module for menu and order management, a User module for browsing and purchasing, and a Delivery dashboard. The backend relies heavily on Firebase Firestore for real-time synchronization. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement a solution for that. It was an overall great experience to have this internship.

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1 Preface

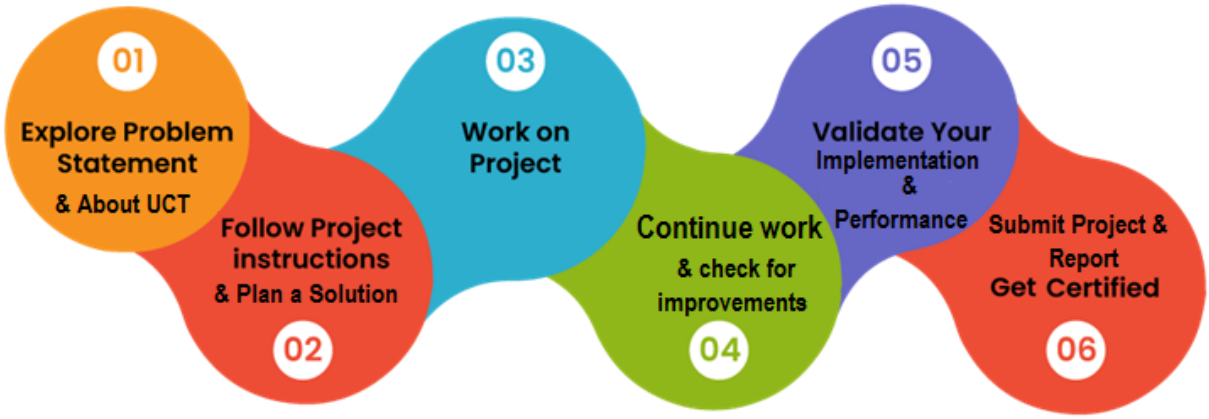
Over the 6-week period, I developed the "TestyBites" Android application. The work encompassed designing XML layouts, implementing Java-based Activities and Adapters, integrating Firebase Authentication and Firestore, and handling real-time map data using OSMDroid.

In software development, particularly mobile app development, theoretical knowledge must be met with practical execution. This internship provided the crucial bridge to understand industry-standard practices, lifecycle management, and cloud architecture deployment.

The project required creating a unified platform where restaurant admins can manage inventory and track live order status, while users can seamlessly browse, add to cart, and process mock payments.

The internship structure provided by USC/UCT offered access to structured learning and a framework to build a production-level application rather than a simple academic prototype.

The program was executed in phases: Exploring requirements, planning the NoSQL schema, developing the User UI, integrating the Admin dashboard, and finally implementing real-time notifications and location tracking.



I gained significant experience in handling asynchronous tasks, data binding using RecyclerView, and managing state transitions based on live database listeners

Thank You to I extend my gratitude to the mentors at upskill Campus and UCT for their guidance and the structured curriculum that made this project possible.

My message to your juniors and peers is to never give up, maintaining consistency is the sole key.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoIe.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

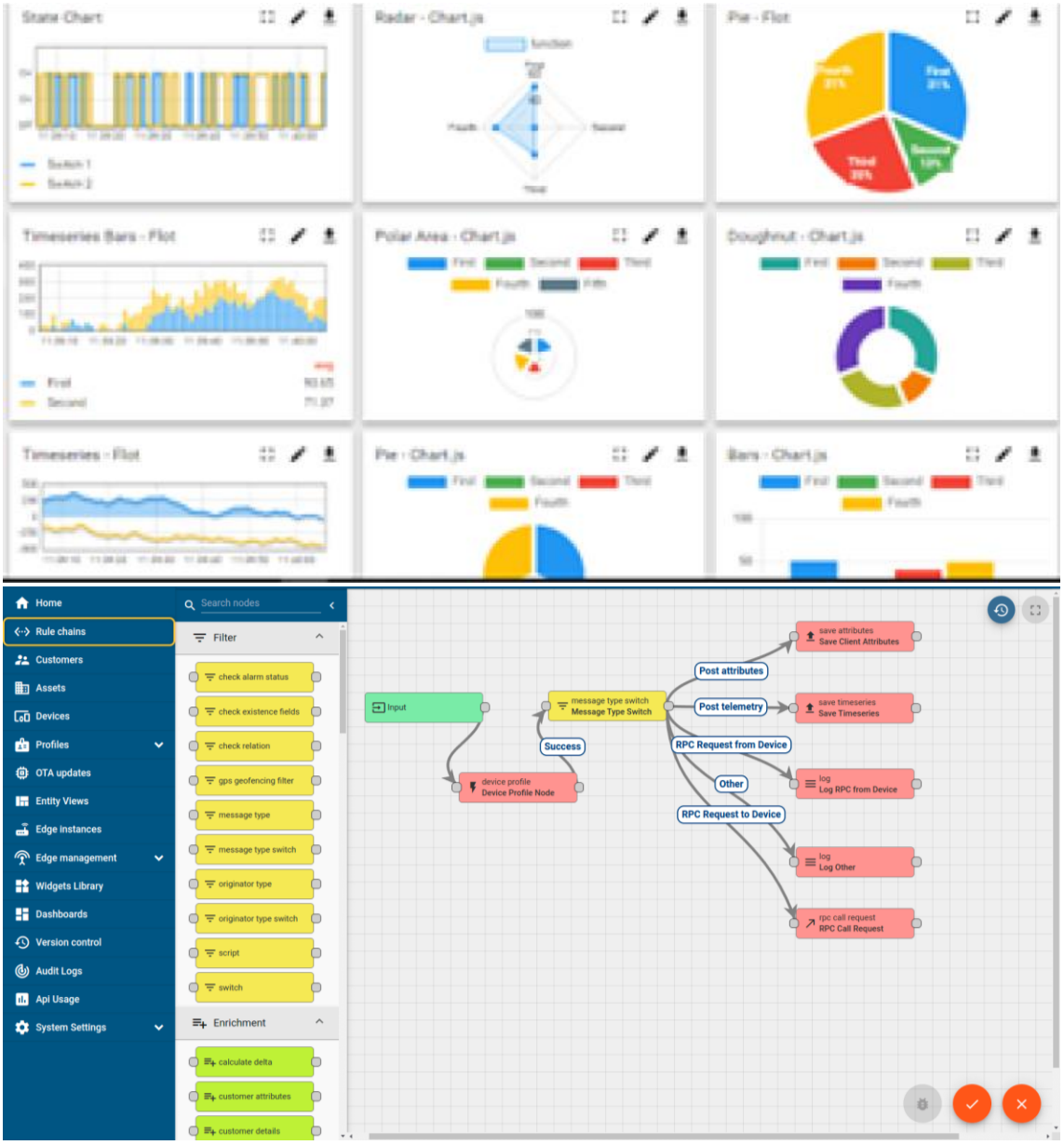
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application (Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- To unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

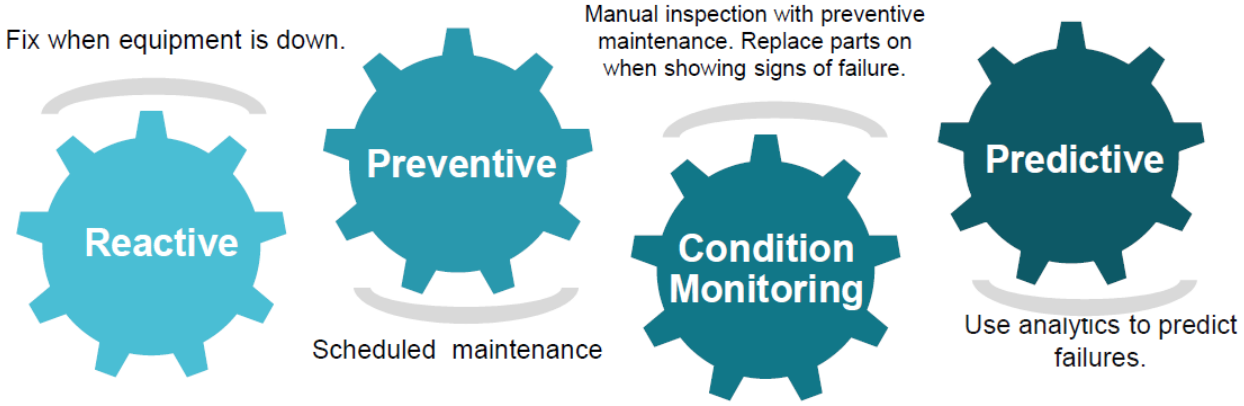
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN Technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

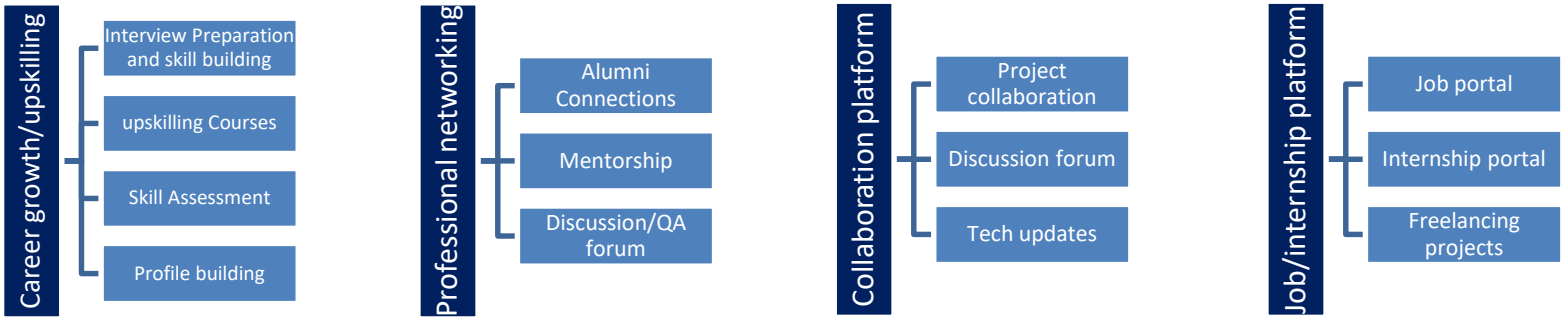
Upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

Seeing need of upskilling in self-paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

UpSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ To solve real world problems.
- ☛ To improve job prospects.
- ☛ To Improved understanding of our field and its applications.
- ☛ For Personal growth like better communication and problem solving.

2.5 Reference

[1] Android Official Documentation (Java).

[2] Firebase Cloud Firestore API Documentation.

[3] Glide Library Implementation Guide.

2.6 Glossary

Terms	Acronym
Firebase	Cloud-based NoSQL database used for app data.
FCM	Firebase Cloud Messaging, a cross-platform messaging solution for reliable notification delivery
RecyclerView	Android UI element used to display scrolling lists.
Glide	Image loading and caching library for Android.
OSMDroid	An open-source library used for integrating map views into Android applications.

3 Problem Statement

In the assigned problem statement, the objective was to develop an Android-based Food Delivery Application that addresses the communication gap between restaurant administrators, customers, and delivery personnel.

The problem required a solution capable of real-time menu management, dynamic cart calculations, secure authentication, and instant status updates without requiring manual app refreshes.

4 Existing and Proposed solution

- **Summary of existing Solutions:** Existing basic solutions often rely on static web forms or phone calls. They lack real-time synchronization, resulting in customers being unaware of their immediate order status ("Cooking", "Out for Delivery") without manually contacting the restaurant.
- **Proposed Solution: TestyBites** is a native Android application deeply integrated with Firebase. It introduces role-based access control (Admin, Driver, User) accessed via specific logins or secret passcodes (e.g., 'admin123'). The application features a dynamic admin dashboard to calculate live revenue, push new food items with images, and instantly update order states. These state changes are immediately pushed to the user's device via Firestore data listeners.
- **Value addition:** The inclusion of dynamic cart calculations, an automated reward point distribution system post-order, and an integrated OSM map view for delivery logistics adds significant real-world application value and industrial readiness.

4.1 Code submission (Github link)

https://github.com/beharemayuri/upskillcampus/blob/main/FoodDeliveryApplication.java

4.2 **Report submission (Github link) :** first make placeholder, copy the link.

https://github.com/beharemayuri/upskillcampus/blob/main/FoodDeliveryApplication_Mayuri_USC_UCT.pdf

5 Proposed Design/ Model

The system design utilizes a Client-Server event-driven architecture. The Android client application relies heavily on the Firebase SDK to communicate asynchronously with the cloud backend for authentication, database reads/writes, and notifications.

5.1 High Level Diagram

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

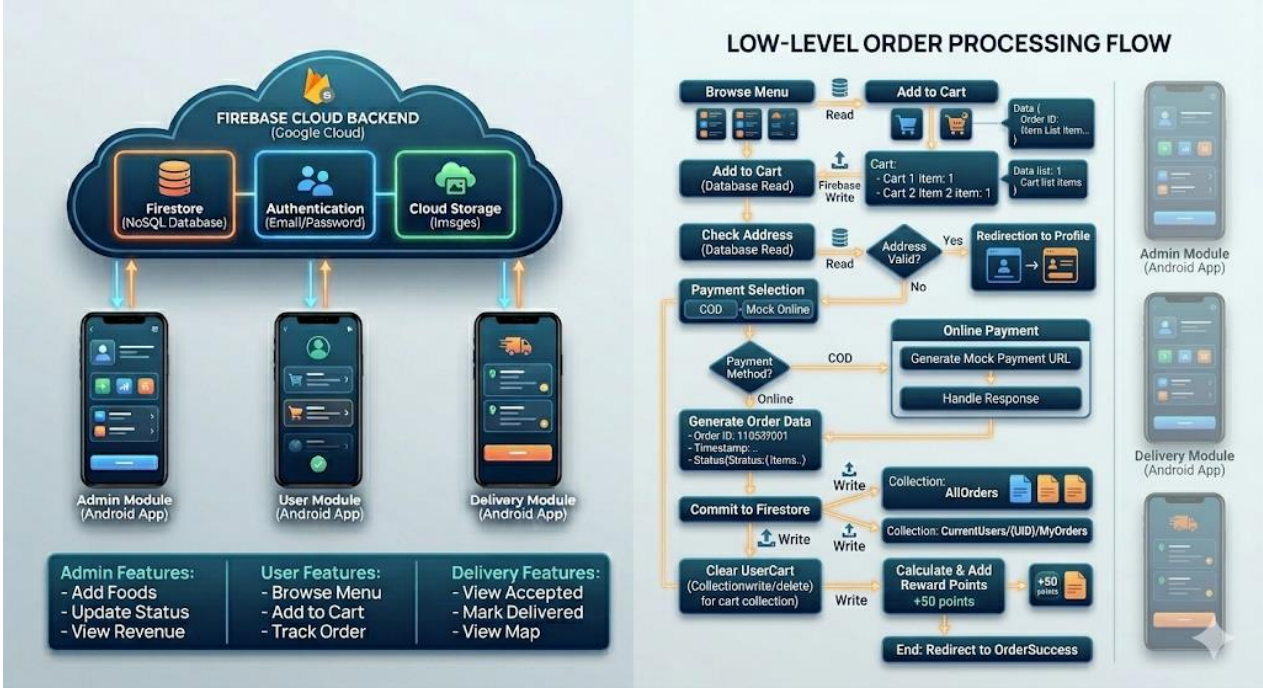


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

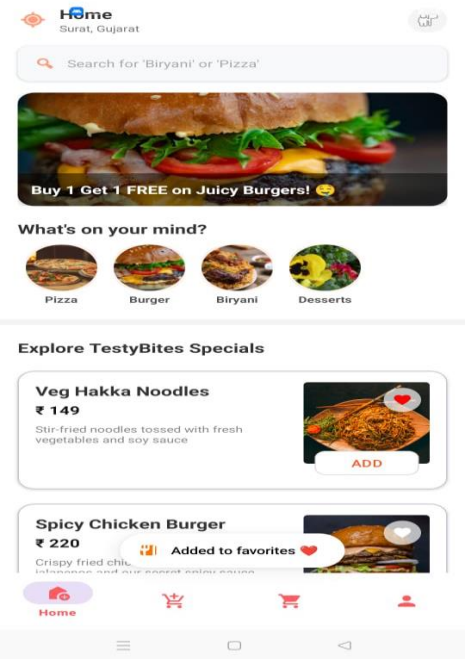
5.3 Interfaces

The application utilizes Android XML for structuring the UI elements.

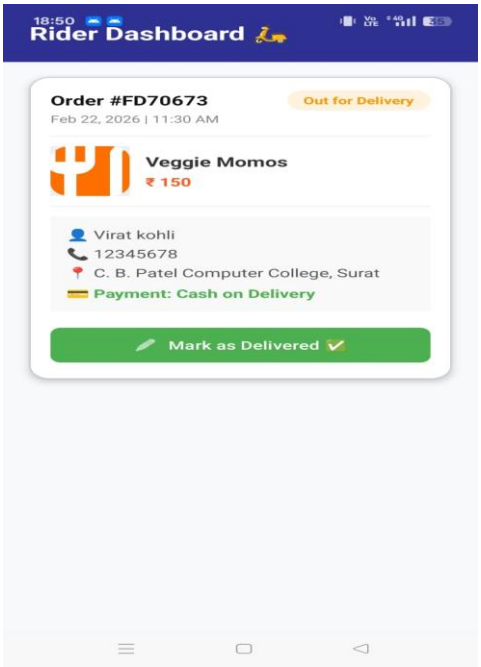
- **Data Flow & Memory Management:** Data payloads are mapped using HashMap<String, Object> and bound to custom Plain Old Java Objects (POJOs) like FoodModel, OrderModel, and CartModel. The Glide library is implemented to manage memory effectively during image loading, preventing OutOfMemory errors when scrolling through large menus.
- **UI Components:** The UI relies on RecyclerView combined with custom Adapters (FoodAdapter, AdminOrderAdapter, CartAdapter) for efficient list management. Material Design components (TextInputEditText, ExtendedFloatingActionButton) ensure a modern, user-friendly interface.

- **Insert a screenshot of the Admin Order Dashboard showing the order status update dialogue here**

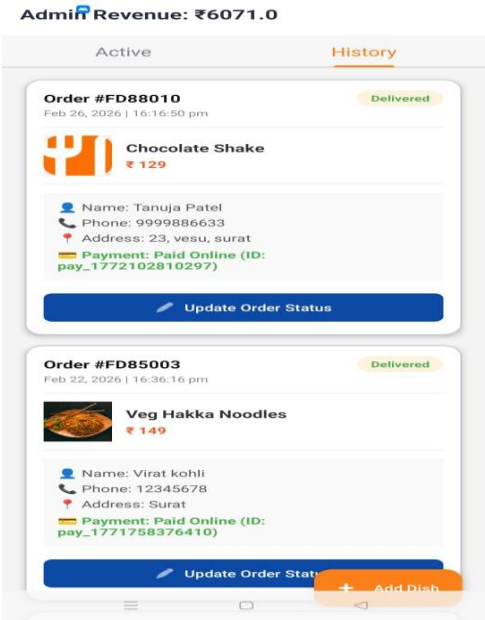
- **UserDashBorad**



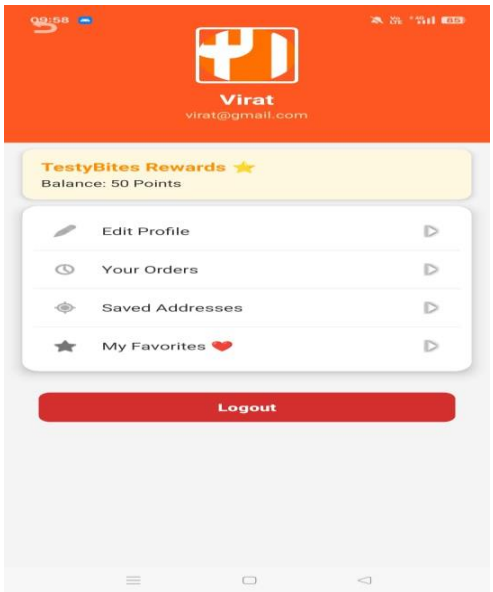
- **Rider Dashboard**



- **AdminPage**



- **User Profile Page**



6 Performance Test

This is a critical section defining why this work translates to real industry application rather than a purely academic project. Testing focused on identifying constraints and ensuring data integrity during asynchronous operations.

6.1 Test Plan/ Test Cases

- **Constraint Identified (Data Formatting):** Null pointer exceptions and Number Format exceptions can occur during cart total calculations if a string price field contains currency symbols or is empty.
- **TC01 (Cart Integrity):** Add multiple items to the cart, increment/decrement quantities, and verify that the Grand Total accurately reflects the (item price * quantity) + delivery fee + taxes.
- **TC02 (Role Access Security):** Attempt to trigger the admin login dialog via tvAdminAccess.setOnLongClickListener and input incorrect access codes. **Expected:** Access denied, UI remains secure.
- **TC03 (Real-time Database Sync):** Change an order status from "Pending" to "Out for Delivery" on the Admin app. **Expected:** The Delivery module list populates the order, and the User app's OrderActivity immediately reflects the new status due to Firestore's real-time capabilities.

6.2 Test Procedure

1. Created a standard user profile, ensuring address fields were populated to pass the validation check (checkAddressAndPlaceOrder()).
2. Added food items and proceeded to checkout using the Mock Payment delay logic to verify that the UI blocking (dialogs) works during processing.
3. Authenticated as an Admin using the secret code to manage the generated orders.
4. Authenticated as a Driver to verify that the DeliveryAdapter filters out "Pending" orders and only displays active deliveries.

6.3 Performance Outcome

The application architecture handled constraints gracefully. The issue regarding price calculation crashes was successfully handled by implementing `try-catch` blocks and regex-based string parsing (`Integer.parseInt(priceStr.replaceAll("[^0-9]", ""))`) to strip non-numeric characters before math operations. The dual-write logic in Firestore during the `placeOrderLoop` executed consistently without data loss.

7 My learnings

This internship provided a profound, hands-on understanding of full-stack mobile application development. I learned how to structure a NoSQL database efficiently to avoid deep nesting and complex queries. I gained practical experience managing the Android Activity lifecycle, utilizing background handlers (`Handler().postDelayed`) for simulated networking, and implementing robust libraries like Glide for image caching and OSMDroid for map rendering. Furthermore, handling asynchronous Firebase callbacks (`addOnSuccessListener`) taught me how to manage UI states effectively to ensure a smooth, crash-free user experience.

8 Future work scope

Due to time limitations within the 6-week program, several features present excellent opportunities for future implementation:

- Integration of a live production payment gateway SDK (e.g., Razorpay or Stripe) to replace the current mock payment dialog.
- Implementing a dynamic GPS tracking API (such as Google Maps Location Services) to show live driver movement on the map interface, replacing the static GeoPoint markers.
- Enhancing the authentication flow to include Google OAuth Sign-In and mobile OTP verification via Firebase Auth.